

Environmental Technology

ISSN: 0959-3330 (Print) 1479-487X (Online) Journal homepage: www.tandfonline.com/journals/tent20

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To cite this article: Priscila Rosseto Camiloti, Freddy Valdés, Tiago Palladino Delforno, Jan Bartacek, Marcelo Zaiat & David Jeison (2019) A membrane aerated biofilm reactor for sulfide control from anaerobically treated wastewater, Environmental Technology, 40:18, 2354-2363, DOI: 10.1080/09593330.2018.1441329

To link to this article: https://doi.org/10.1080/09593330.2018.1441329

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Published online: 26 Feb 2018.

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ENVIRONMENTAL TECHNOLOGY
2019, VOL. 40, NO. 18, 2354–2363
https://doi.org/10.1080/09593330.2018.1441329

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A membrane aerated biofilm reactor for sulfide control from anaerobically treated wastewater

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ABSTRACT

A upflow anaerobic sludge blanket reactor was operated combined to a membrane aerated biofilm reactor for sulfate removal and for elemental sulfur reclamation. A commercial silicon tube was used as an oxygen delivery diffuser. The process achieved high rates of sulfide removal from the liquid phase (90%). The hydrogen sulfide removal was influenced by the pH value and at pH value of 7.5, 98% of the H_2S was removed. The elemental sulfur was observed inside the membrane, with content in the biomass of 21%. Through the massive sequencing of the samples, the microbial community diversity and the stratification of biomass inside the silicon tube was demonstrated, confirming the presence of sulfide-oxidizing bacteria on the membrane wall. The most important genera found related to the sulfur cycle were *Sulfuricurvum, Geovibrio*, *Flexispira* and *Sulforospirillum*.

ARTICLE HISTORY
Received 22 September 2017
Accepted 10 February 2018

KEYWORDS
Sulfide oxidation; elemental sulfur; microaeration; silicon membrane; high-throughput sequencing

Highlights

- Dissolved and gaseous sulfide removal achieved >90% in microaerobic conditions.
- Gaseous sulfide removal was influenced by the pH value; 98% at pH value of 7.5.
- The elemental sulfur was observed inside the silicone membrane.
- Microbial stratification was observed in different layers inside the membrane.
- *Sulfuricurvum, Geovibrio*, *Flexispira* and *Sulforospirillum* were sulfur cycle genera.

1. Introduction

Anaerobic wastewater treatment systems generate waste streams rich in hydrogen sulfide in the presence of sulfate. Because their negative impact in many ways, such as a strong odor, corrosion of steel and concrete structures and health hazards, waste streams must be treated. The biogas quality obtained from anaerobic processes is another example of the need for removing sulfide, as hydrogen sulfide is very corrosive to some parts of gas motors and its combustion produces SO_x emission.

Hydrogen sulfide could be removed by physicochemical and biological methods. Among the biological methods, the microaeration has grown, since is reliable, simple and economically efficient method [1] and consists of applying controlled doses of oxygen in order to promote the activity of sulfur-oxidizing bacteria (SOB). The microaeration for sulfide removal has been reported by Buisman et al. [2] and Jansen et al. [3], and consists of applying controlled doses of oxygen in order to promote the activity of SOB.

Tang et al. [4] reported that the kinetics of sulfide removal by a suspension of SOB biomass confirmed that the final product of the oxidation of sulfides depends on the proportion of dissolved oxygen and concentration of sulfide sulfur. If the oxygen dose is adequately controlled, chemolithotrophic SOB are capable of oxidizing sulfur compounds to oxidation state 0 and show the feasibility of the treatment while preserving the methane production. [5–7]. The microaeration mechanism for sulfide removal involves the formation of elemental sulfur, an insoluble intermediate that can be separated from the liquid phase and reused [8], which represents other advantage to the treatment process. Krayzelova et al. [1] summarize in a review paper some methods to use the microaeration, which refer to controlled dosing of a

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Supplemental data for this article can be accessed at https://doi.org/10.1080/09593330.2018.1441329
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small amount of air/oxygen into the liquid or gaseous phase of anaerobic digesters. The direct air/oxygen injection in the liquid phase or separated vessels is used in most reactor configurations [9–12] and the sulfide removal in the biogas was observed.

The oxygen transfer could be a limiting step during H_2S removal from biogas in microaerobic reactors [13], and a bubbleless oxygenation could be an alternative to controlled and is the application of membranes to the aeration. MABRs are considered a new technology for aerobic wastewater treatment. Oxygen diffuses through the membrane into the biofilm where oxidation occurs whereby the biomass is immobilized on membranes through which oxygen is supplied. [14]. Using membranes in tubes offers additional benefits such as a large surface area of biofilm per unit volume of bioreactor and the ability to operate at very high flow rates over the biofilm in order to enhance substrate transfer into the biofilm [15]. Considering this, a silicone rubber is a potentially efficient means for supplying oxygen via diffusion through bubbleless gas transfer. The oxygen is transported through the membrane and goes directly into the liquid phase in a dissolved form, because of the high contact time, achieving up to 100% of oxygen transfers [16]. This characteristic would imply a minimal oxygen transfer from the liquid phase to biogas, which results in a great advantage over the direct microaeration in the headspace of the digester, and the traditional aeration in the liquid phase.

In recent years, there has been increasing interest in membrane bioreactors for wastewater treatment [17] and its use has been reported for nitrogen removal [18–20], simultaneous nitrification and organic carbon removal [21–23]. Using silicon membranes has been also studied in membrane aerated biofilm reactor (MABR) for acetate removal [15], food processing wastewater [24], phenol [25] and chlorophenols [26]. Conversely, there is a lack of information about the use of this system in sulfide removal.

The aim of this study is to evaluate the feasibility of a combined upflow anaerobic sludge blanket (UASB)–MABR using silicon membrane for treating sulfate and sulfide containing wastewater. Moreover, the other objectives of this study were to explore the sulfide conversion to sulfur, the influence of the pH value on the dissolved and gaseous sulfide removal and analyze the microbial community structure and function of the dominant species in the biomass inside the membrane.

2. Materials and methods

2.1. Bioreactor configuration

The system used in this study was composed of an UASB reactor connected with a MABR, which consisted of a silicon membrane inside a chamber. A schematic diagram of the UASB–MABR system is shown in Figure 1.

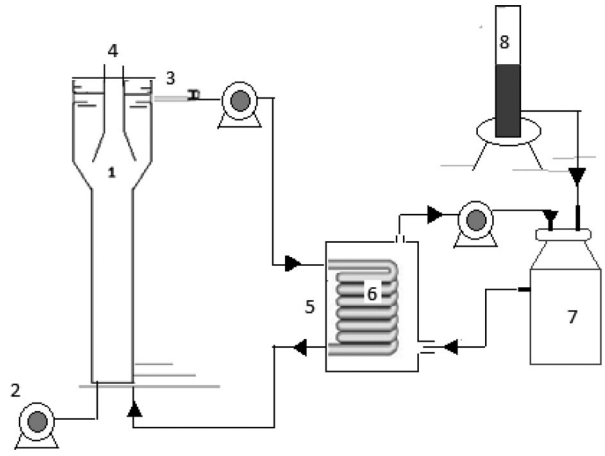


Figure 1. Schematic diagram of the anaerobic/microaerobic bioreactor system. 1- UASB reactor; 2- feed pump; 3- effluent; 4- biogas out; 5- MABR; 6- silicon membrane; 7-oxygem bottle; 8- graduated cylinder.

The UASB reactor (Figure 1-1) was fed with a flow rate of 2.1 L d^{-1}. Effluent from the UASB reactor was conducted into the MABR (Figure 1-5) for sequenced biological sulfide oxidation. The effluent was circulated through the silicon membrane by a positive displacement pump with a flow of 2.7 L h^{-1}.

The silicon membrane (Figure 1-6) was connected directly to the UASB reactor and was maintained inside a chamber (Figure 1-5). The chamber was connected to a bottle, with a working volume of 2.0 L. which was filled with pure oxygen under constant pressure. The oxygen stored in the bottle was pumping to the chamber with a flow of 2.4 L h^{-1}. The bottle was used in the system to feed the MABR and to allow the gas sampling to observe biogas diffusion through the membrane.

The UASB reactor has an effective volume of 2.2 L and was seeded with granular sludge obtained from a full-scale UASB plant in a brewery (CCU Cervecera, Temuco, Chile).

The silicon membrane had an internal diameter of 8.0 mm, a wall thickness of 2.4 mm and 2000 mm long. A single silicon tube was used to allow the biofilm development and was maintained throughout the reactor operation inside a chamber to permit the measure the gas transfer.

2.2. Operational conditions

Three reactors were operated in this study. Two reactors were operated for 100 days. An UASB reactor was operated without microaeration for control and the UASB–MABR system (1) was operated with microaeration. The UASB–MABR system (2) was also operated for 78 days

Table 1. Operational conditions during the study.

Reactor_ID	Stages	Condition	pH	Operation time (days)
UASB Control	–	Anaerobic	7.0	101
UASB_MABR 1	1	Anaerobic	7.0	20
	2	Micro aerobic	7.0	30
	3	Micro aerobic	7.0	59
UASB_MABR 2	1	Anaerobic	7.5	23
	2	Micro aerobic	7.5	43
	3	Micro aerobic	7.0	16

to determine the influence of pH value (pH 7.0 and 7.5). The temperature was maintained at 30 ± 1°C.

The feed consisted of diluted wine supplemented with anhydrous sodium sulfate to increase the amount of sulfide and sodium bicarbonate (10 g L^{-1}). The volumetric loading rates were approximately 6.0 kgCOD L^{-1} d^{-1} and 0.1 kgSO_4^{2-} L^{-1} d^{-1}. Sulfur and chemical oxygen demand (COD) balances were monitored regularly by measuring the flows of influent and biogas and analyzing liquid and gas samples. Table 1 summarizes the periods of the study.

The operation of the UASB–MABR system was divided into three stages. Stage I, before starting the microaeration (day 0–20). Stage II, during the microaeration until the removal decrease (day 20–49). Stage III, silicon membrane occlusion which was characterized by the sharp drop in sulfide removal (day 51–109).

2.3. Biofilm start-up

The biofilm start-up took approximately 7 days to develop the biofilm from inoculation to a visual biofilm formation. The start-up biofilm growth inside the tube was encouraged by circulating through membrane tube synthetic sewage under an organic loading rate of 3.4 kgCOD L^{-1} d^{-1}. On the second day of operation, a thin film could be observed on the membrane wall (MW) and on the sixth day of operation, a black biomass above the thin film was observed. The rate of biofilm thickness increase could not be observed due to system impossibility opening. The biofilm formed inside the tube was collected at the end of the operation on the 109th day. Furthermore, the total solids and volatile suspended solids (VSS) were determined.

2.4. Batch tests

The MABR was disconnected from the system on the 92nd day and connected to a bottle to perform batch experiments in order to observe the capacity of the anaerobic biofilm layer to reduce the sulfur to sulfide. The batch experiments were performed in 500 mL glass bottles filled with 500 mL of basal medium containing 1 g L^{-1} of $NaHCO_3$. The bottle was incubated at 30°C on a shaker at 50 rpm. Ethanol was added after 210 min (0.5 gCOD L^{-1}) as an electron donor to the sulfur reduction. Sulfide, sulfate, thiosulfate and COD were measured in selected time intervals.

2.5. Physical–chemical analysis

Liquid samples were collected from influent and effluent from the UASB reactors. The sulfate and thiosulfate ions were analyzed by ionic chromatography (Metrohm, model 882). Samples were diluted 1:4 with a 0.1 M zinc acetate solution, followed by centrifugation (5 min at 16,000 g), thus precipitating sulfide as zinc sulfide [27]. The supernatant was filtered at 0.22 µm and the sample was injected. Sample separation and elution were performed by an Metrosep A Supp 4 – 250/4.0 analytical column using carbonate/bicarbonate eluent (4.5 mmol Na_2CO_3/0.8 mmol $NaHCO_3$ at 1 mL min^{-1}). Dissolved sulfide was measured by the iodometric method specified in Standard Methods [28]. Sulfur content in the sludge and in the biofilm was estimated from an elemental analysis of sulfur. Samples were dried at 95°C, cooled and analyzed in an elemental analyzer (euroEA 3000- Eurovector). COD and VSS were determined according to standard methods [28]. The biogas production was measured by liquid displacement in an inverted cylinder. Biogas composition (O_2, N_2, CO_2, CH_4) was measured by gas chromatography (GC-TCD) Clarus 580 (PerkinElmer) equipped with a thermal conductivity detector. A Carboxen 1000 (15 ft × 1/8 in ss) column was used. Helium was the carrier gas. The inlet, detector and oven temperature were 200°C, 250°C and 120°C, respectively. The H_2S in biogas was analyzed using colorimetric gas detection tubes (RAE system). The pH values were monitored online using a specific probe and adjusted by HCl or NaOH (1 M) with a flow of 0.4 mL min^{-1}.

2.6. Biological analysis by 16s rRNA gene sequencing

A massive amplicon sequencing of the rRNA 16S on a Miseq 2 × 250 – Illumina platform, (Illumina Inc., San Diego, CA, U.S.A.) was carried out to identify the microbial community in the reactor. Three samples were collected at the end of reactor operation from MABR to evaluate the biological diversity in each sample. The samples were collected from three different points: the sample from the MW; from sulfur formations attached to the MW (SF) and from the middle of the membrane – the internal biomass (IB). The sites of sample collection were chosen based on physical differences noted in the biofilm.

The collected samples were retrieved by successive washing in phosphate-buffered and subsequent

centrifugation. The extraction of total DNA was made using the phenol: chloroform-based protocol described by [29]. DNA quality was assessed by the 260/280-nm >1.8, measured by an ND-2000 spectrophotometer (Nanodrop Inc., Wilmington, DE, U.S.A.). Furthermore, rRNA genes were amplified for massive sequencing analysis using a primer set that flanked the V3 and V4 hypervariable region of the 16S rRNA gene [30]. The full-length primer sequences, using standard IUPAC nucleotide nomenclature, to follow the protocol targeting this region are: 16S Amplicon PCR Forward Primer primers 16S (S-D-Bact-0341-b-S-17) (CCTACGGGNGGCWGCAG) and 16S Amplicon PCR Reverse Primer S-D-Bact-0785-a-A-21 (GACTACHVGGGTATCTAATCC).

The sequences were pre-processed using the CASAVA (1.8.2) Software. Sequences were first trimmed to remove the adaptor, barcodes, primers and sequences containing ambiguous 'N' or shorter than 200 bps (Software Seqyclean (https://bitbucket.org/izhbannikov/seqyclean). The minimum read Q score adopted was 24 (Phred Quality). Sequences shorter than 65 bp, or containing ambiguous nucleotides, or with an occurrence lower or equal to 65 were excluded using the Seqyclean program. The vector contamination was removed with the UNIVEC contaminant base (http://www.ncbi.nlm.nih.gov/VecScreen/UniVec.html).

The alignment of the sequences was performed using the Pynast tool according to Caporaso et al. [31]. To determine the operational taxonomic units (OTU), 'hierarchical clustering' was used with 97% similarity. Singleton sequences (OTU with one sequence) that may represent sequencing errors [32] were removed.

For the taxonomic classification of sequences representative of each OTUs, the RDP-Classifier was used. The confidence threshold adopted was 80% for genus and 50% for other taxonomic levels (Phylum-Family). Alfa (Chao1, Shannon, Simpson and Dominance) and Beta (Bray–Curtis dissimilarity index) diversity were quantified using Past software [33]. For this, the dataset was normalized using the QIIME script [31]. The Circos diagram [34] was used to compare and visualize the genus found from the samples.

The sequences generated in this study were deposited in the European Nucleotide Archive (http://www.ebi.ac.uk) under project accession number is PRJEB12845.

3. Results and discussion

3.1. Performance of the UASB–MABR system

After starting the microaeration until the removal decrease (stage 2 – day 20–49), the reactor produced a biogas with an average of hydrogen sulfide concentration and average removal in the biogas of 377.8 ± 156.3 ppmv and 98%, respectively. The dissolved sulfide had an average removal and concentration of 90% and 10.9 ± 5.8 mg L^{-1}. During this period, the hydrogen sulfide concentration was below 400 ppmv in 8 out 11 samples. The sulfate and COD removal was stable during the whole operation showing an average of 69.9 ± 11.8% and 88.0 ± 3.7%, respectively. The dissolved sulfide and the sulfide in the biogas concentrations in the UASB–MABR system are shown in Figure 1.

Organic matter removal and the methane production were not significantly affected by microaeration. The COD removal shows an average of 88.0 ± 3.7% and 85.0 ± 7.6% for the UASB–MABR system and for the control, respectively. The concentration of methane and carbon dioxide in the biogas in the UASB–MABR system showed the same behavior as the control reactor. The average methane concentration in % (v/v) was 54.2 ± 14.7 and 52.2 ± 14.4 and the average carbon dioxide was 34.2 ± 11.7 and 35.7 ± 22.5 for the UASB–MABR system and the control, respectively. This indicates that limited oxygen supply did not reduce the energetic content of the biogas.

The biogas that diffused through the membrane was analyzed and the sampling was performed on the chamber. The concentration of the main biogas components, methane and carbon dioxide, showed an average of 23.1 ± 14.9 and 23.2 ± 8.8% (v/v), respectively. During the biofilm start-up (day 20–27) the sulfide concentration in the chamber showed an average of 30.0 ± 11.9 mg L^{-1} and after the biofilm formation the sulfide concentration was not detected (day 29–65), demonstrating the biofilm capacity of sulfide removal.

In stage 3 (day 51–109), the concentration of dissolved sulfide and sulfide in the biogas increased, showing an average concentration of 23.9 ± 6.4 mg L^{-1} and 2560 ± 1030 ppmv (Figure 2). In this phase, the biofilm became thick, which makes it difficult to transfer oxygen.

The concept of different layers in the biofilm is important to understand the performance of the UASB–MABR system. Casey et al. [14] report that in the conventional biofilms, the active layer is generally an oxic layer, typically 50–200 mm in thickness, which corresponds to the depth of oxygen penetration into the biofilm. If the overall biofilm thickness is greater than the oxygen penetration depth, then an anoxic layer, adjacent to the biofilm support, occurs. The different layers occur in this study. The layer attached to the inner surface of the membrane was responsible for the sulfide oxidation; however, the increase in biofilm thickness may have caused a resistance to the sulfide reaction with oxygen.

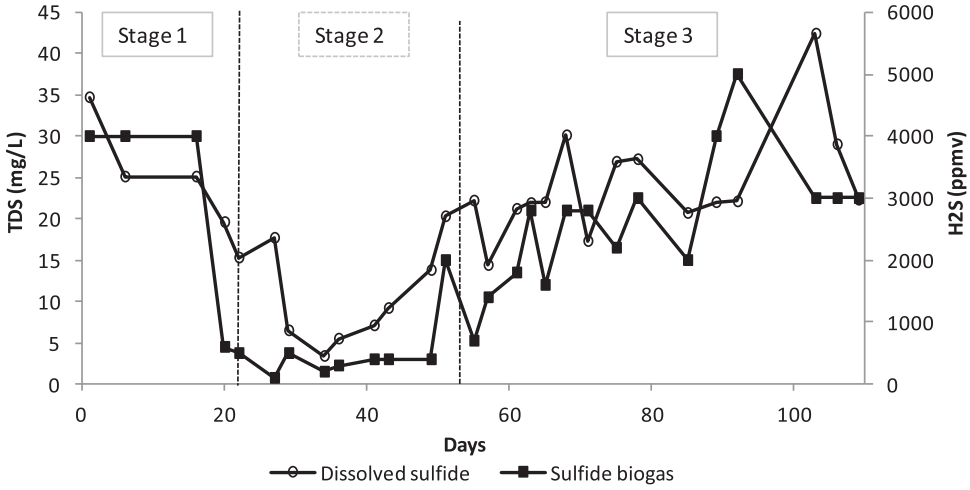


Figure 2. Concentration of total dissolved sulfide (TDS) in the effluent and hydrogen sulfide in biogas.

At stage 3, the elemental sulfur reduction to sulfide could be proved by the batch test with ethanol addition (Figure 3). It can be observed that there was an increase in the sulfide concentration accompanied by the decrease in ethanol and dissolved oxygen concentration, indicating the reduction of sulfur to sulfide. At the beginning of the operation, a concentration of 550 μg L^{-1} of sulfide was measured, possibly by the biofilm first wash. The sulfide concentration has a decrease after 30 min, with an average concentration of 59.2 ± 19.5 μg L^{-1}. After 240 min, ethanol was added and the sulfide concentration showed an increase. The dissolved oxygen also showed a decrease after 30 min, however, the concentration showed a slight decrease, probably by the oxidation of ethanol. The oxygen utilization to the sulfide or sulfur oxidation did not occur as sulfate and thiosulfate was not observed.

Similar reactors using immobilized microorganisms on membranes used to the aeration have also demonstrated the decline in the efficiency with the biofilm increase. Pankhania et al. [35] reported a decline in the COD efficiency in a reactor using hollow fiber and used the backwash to reduce the biofilm thickness. The authors also concluded that the decline in COD removal efficiency prior to the backwash could be due to reduced mass transfer as a result of increasing biofilm thickness or declining biomass activity with metabolite accumulation in the dense biofilm. Casey et al. [15] also reported that an optimum biofilm thickness exists, which maximizes the overall reaction rate by minimizing the diffusional resistance of substrate and nutrient transfer in sections of the biofilm where biodegradation does not take place. Using a tubular silicon membrane, where the microorganisms grow inside the membrane, the oxygen mass transfer will occur in the MW and the active layer to the sulfide conversion is close to the biofilm–membrane interface. Unless the biofilm thickness is the same as the thickness of the active layer, a diffusional resistance to oxygen and sulfide transfer through the biofilm will occur [15]. In the present study, the thick biofilm did not completely penetrate the oxygen and sulfide in the biofilm and dual limitation predominated. In this case, the optimization and stable sulfide removal efficiency can easily be achieved by backwashing the membrane.

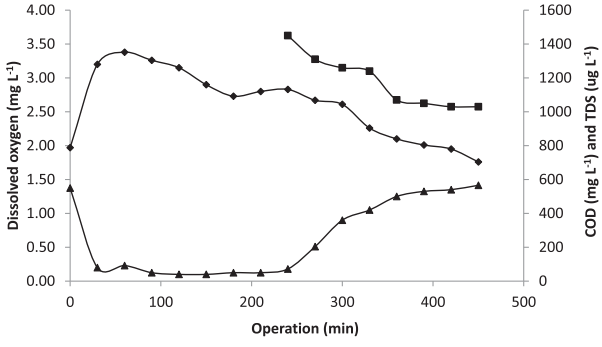


Figure 3. Sulfide production in a batch experiment in the presence of ethanol. (▲) sulfide; (♦) dissolved oxygen; (■) COD.

3.2. Sulfur balance

A sulfur balance was performed in order to study the results of sulfide oxidation and to investigate its conversion to sulfur in the UASB–MABR system. The UASB–MABR system and reactor control performance data, in terms of S-containing compounds, are presented in Table 2. During stage 2, the introduction of oxygen substantially altered the sulfur outputs of the process. From the effluent data, it can be observed that the complete re-oxidation to sulfate did not occur and neither did

Table 2. Sulfur mass balance during continuous operational stages of the anaerobic control reactor and the UASB–MABR system.

		Influent (mgS L^{-1} dia^{-1})	Effluent (mgS L^{-1} dia^{-1})			Balance (%)	
		$S–SO_4^{2-}$	$S–SO_4^{2--}$	$S–SO^{2-}_{-effl.}$	$S–SO^{2-}_{-biog}$	Total	Gap
Control	–	128.2 ± 33.5	47.1 ± 7.8	50.7 ± 8.7	10.2 ± 6.3	89.0	11.0
UASB/MABR 1	Stage 1	94.0 ± 27.7	40.2 ± 9.1	47.9 ± 9.4	21.8 ± 4.7	91.0	9.0
	Stage 2	123.1 ± 9.9	47.1 ± 9.7	20.1 ± 10.6	2.7 ± 3.1	53.5	46.5
	Stage 3	165.0 ± 18.2	43.9 ± 7.6	41.1 ± 11.4	6.2 ± 6.2	66.0	34.0

oxidation to thiosulfate or sulfite. The gap in the S-balance, 46% and 34% for stage 2 and 3, respectively, may have been the fraction of elemental sulfur.

In the control and anaerobic phase of the UASB–MABR system (stage 1), 89% and 91% of the S influent could be recovered as sulfate-S, sulfide-S effluent and in biogas. During stage 2, elemental sulfur was observed and was maintained attached in the membrane. However, since the membrane chamber remains sealed throughout the operation, it was not possible to assess this fraction reliably. The amount of sulfur transformed was 53.5% and 60.0% for stage 2 and 3, respectively, with a gap of 46.5% and 34.0%, this gap is related to the formation of elemental sulfur in microaerobic conditions. After the operation, biofilm formed inside the silicone membrane was taken and the elemental analysis showed sulfur content for biofilm equal to 21.0%.

Probably the higher average concentration of dissolved sulfide and hydrogen sulfide during stage 3 compared to stage 2, is the reason for the lower gap in the absolute value in stage 3 compared to stage 2. While in stage 2, sulfide was removed leading to a gap around 46.5%, in stage 3, the concentrations of sulfide increase, leading to a gap around 34%. The difference in the gap in the sulfur balance, and consequently the decrease in the sulfide conversion to sulfur, could be due to the biofilm growth and the additional resistance in the biomass to the sulfide reaction with the oxygen.

3.3. pH effect

The pH effect was also studied simultaneously in another system for 82 days. The system was also a UASB–MABR (2) with the same conditions and the biofilm developed was similar to the other one. The pH value significantly affects the undissociated H_2S form of sulfide which, by the displacement of the equilibrium, leads the sulfide from the gas phase to the liquid phase, which is further treated in the microaerobic system. For the pH value of 7.5 (stage 2), the sulfide concentration in biogas had a maximum removal of 98% and for pH 7.0 (stage 3), the maximum removal was 50%. Figure 4 shows the average and standard deviation to dissolve the sulfide and hydrogen sulfide concentration for each stage.

The microaeration combined with alkaline conditions was beneficial. While microaeration with air dosed into the gas phase removed only hydrogen sulfide from biogas, others studies with injection directly into the liquid could remove hydrogen sulfide from biogas and liquid phase. However, the efficiency of hydrogen removal was not high in this study (stage 3) and others studies with low concentrations of oxygen [9].

Summary, our findings about the use of the UASB–MABR for the sulfide conversion to the elemental sulfur indicate that the oxygen supply through the silicon membrane combined with alkaline pH promote the sulfide removal both for the liquid and gas phase. Therefore, the additional step of removing hydrogen sulfide from

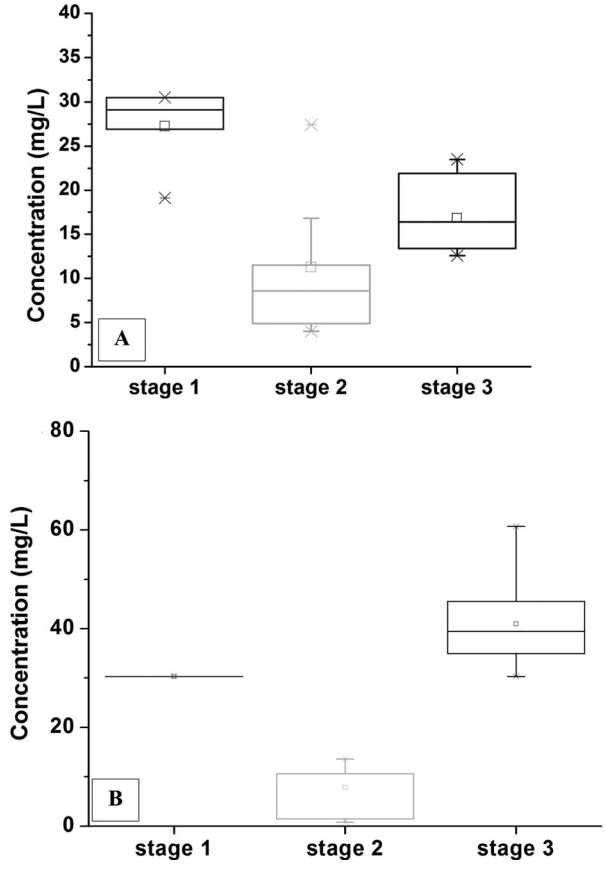


Figure 4. Dissolved sulfide (a) and hydrogen sulfide in biogas (b) during the anaerobic stage (stage1), pH value of 7.5 (stage 2) and pH value of 7.0 (stage 3).

the biogas is not necessary, as the sulfide in biogas could be transferred to the liquid phase and be removed.

3.4. Microbiological analysis

After opening the system, the biofilm could be analyzed and was observed different layers, which have developed and differentiated naturally. The biofilm stratification could be observed visually. In this way, the composition of the bacterial community in the three different layers was studied and showed that different communities were established throughout the biofilm. The samples were taken from biomass attached to the silicon MW, from the elemental sulfur formation in the membrane (SF) and from the middle of the membrane (IB). The microorganisms related to the sulfide oxidation were found attached to the inner surface of the membrane and the sample taken from the middle of the membrane mostly showed an anaerobic community.

In total, out of the three samples, 76,128–149,693 raw sequences were generated with an average length of 394 ± 92 bp. During the processing of sequences, 40–47% of sequences were removed due to the low quality of the bases (Qphred > 24), lengths less than 200 bp, chimeras or ambiguous bases. The values of coverage estimation (Good's formula) ranged from 98% to 99%, indicating a high coverage of the diversity.

The analyses of the rarefaction curves were used to infer that the phylum taxonomic level (80% similarity – phylum level) and the number of sequences obtained were sufficient to represent all the diversity in the biomass (Supplementary Figure 1). However, the rarefaction curves at 95% (genus level) and 97% (species level) of similarity showed that more sequence is necessary to access the full diversity of genera and species. Furthermore, the richness estimator (Chao 1) values observed in the MW sample (5057 ± 173) were higher than those observed in the SF sample (2209 ± 119) and IB sample (832 ± 68). The Shannon diversity index yielded values greater than 4.0 for all samples, which shows high diversity (MW: 4.64 ± 0.02; SF: 4.22 ± 0.03 and IB: 4.86 ± 0.01).

By using a Bray–Curtis similarity dendrogram (Figure S1(a)), the similarities among the communities were determined and show a high similarity (64%) between the sulfur formation and the sample drawn from the MW. A similarity of 42% was observed between the samples taken from the MW (MW and SF) and the sample drawn from the IB. The clustering of MW samples may be due to the microaerobic conditions and the low similarity between the IB sample and the MW and SF samples indicate the stratification layers of the biofilm. In addition, the Venn diagram showed that 23% of OTUs were exclusive for the samples taken from the MW (13% for MW and 10% for SF), while the IB sample had 7.5% exclusive OTUs (Figure S1(b)).

To understand the microbial community structure and the biological stratification in the UASB–MABR system at the three stages, the classified OTUs were analyzed at phylum and family level (Figure 5). Meanwhile, to explain the key role of the dominant genera, the classified OTUs were analyzed at the genus level (Figure 6). The RDP-Classifier was used for the taxonomic classification and showed that over 96% of the sequences were classified at the phylum level. At the genera level, the percentage ranged from 53% to 68% of the classified sequences. The main phyla founded for the samples MW, SF and IB (Figure 5) were the *Proteobacteria* (54.1%; 64.9% and 28.5%), *Firmicutes* (7.9%; 8.2% and 24.2%) and the *Bacteriodetes* phylum (20.2%; 14.1% and 24.9%).

In total, for the three samples, 19 genera were identified. The most abundant genus was the *Sulfuricurvum* genera. In addition, six other genera related to the sulfur cycle were found. Two of them are related to oxidation of the sulfide and were exclusively found in samples withdrawn from the MW, which were *Geovibrio* (3.2% for MW and 2.4% for SF), *Flexispira* (1.2% for MW and 2.8% for SF) and *Sulfurospirillum*. (1.1–1.2% for MW and SF). It is also important to emphasize the role of sulfate-reducing bacteria, which can be attached to *Paludibacter, Desulfovibrio* and *Desulfobulbus* genera and

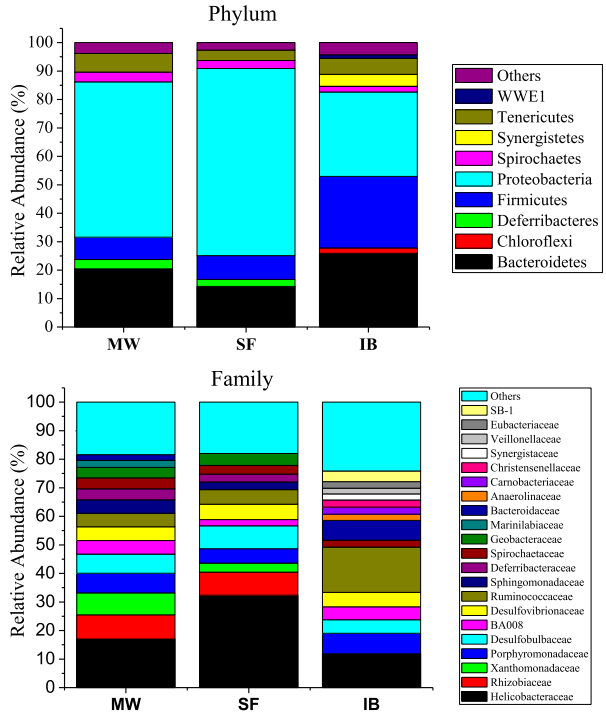


Figure 5. Microbial community structure of the samples at the phylum and family level.

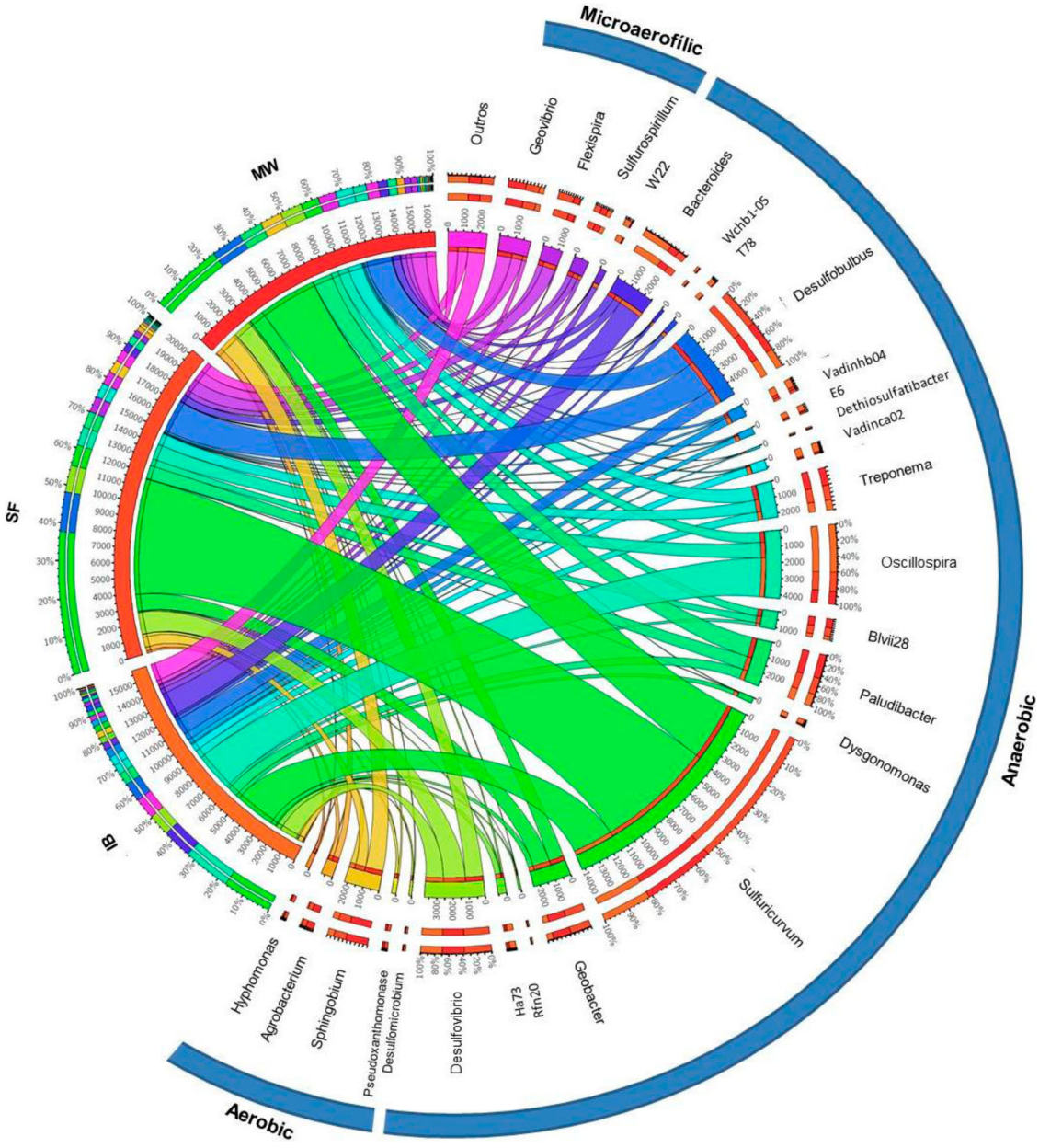


Figure 6. The main classified genera using the Circos diagram and divided into anaerobic, aerobic and microaerobic groups. Samples taken from the MW, (B) sulfur formations attached to the MW – SF and IB.

were found in three samples with relative abundance ranging from 0.4% to 7.8%.

Figure 6 shows the Circos diagram, which lists the samples to the genera found in each sample and the relative abundance. In addition, the genera were divided into three groups: anaerobic, aerobic and microaerobic. Among the major genera, *Sulfuricurvum* genera represented more than 24.0% and 37.9% (relative abundance) of the MW and SF sample and 16.8% for the IB sample. The species related to the genera was *Sulfuricurvum kujiense* and was found in the three samples. The species was described by Kodama and Watanabe [36] as a facultative anaerobic sulfur-oxidizing bacterium that grows anaerobically and microaerobically by oxidizing reduced sulfur species, such as sulfide, elemental sulfur and thiosulfate, however, under anaerobic conditions, nitrate can be used as an electron acceptor.

It can also be observed in the Circos diagram that all the samples are related to the three groups, but sample MW shows most genera associated to the anaerobic group. In the anaerobic group, the following sulfate-reducing bacteria can be observed: *Desulfobulbus, Desulfovibrio* and *Paludibacter*. Another anaerobic species with high relative abundance was the *Geobacter lovleyi* species. This species was identified in the three samples, but with greater abundance in the samples

taken from the wall membrane samples. Although anaerobic species were found in the samples attached to the membrane, this species can be associated to using elemental sulfur as an electron acceptor [37], which reduces sulfur and sulfide production.

The four microaerobic genera found in the greatest abundance are related to the sulfur cycle. Moreover, *Sulfuricurvum, Geovibrio*, *Flexispira* and *Sulforospirillum* were found, and the last three were exclusively found in samples taken from the MW (MW and SF samples), indicating the stratification and development of the biomass of different communities. The genus *Geovibrio* is related to oxidation of the sulfide [38], although the *Flexispira* and *Sulforospirillum* are associated to the reduction of elemental sulfur.

Finally, through the massive sequencing of the samples, the stratification of the biomass inside the silicon tube was confirmed. The development of microorganisms related to oxidation of sulfide in the MW confirms the hypothesis that the mass transfer resistance may have provided different genera, resulting in different layers inside the tube.

4. Conclusions

The dissolved and gaseous sulfide removal achieved, in microaerobic conditions, efficiency higher than 90% and 98%, respectively. The gaseous sulfide removal was influenced by the pH value, at a pH value of 7.5, 98% of the H_2S was removed from biogas and further treated in the microaerobic reactor. Thus, the oxygen supply through the silicon membrane combined with alkaline pH promotes the sulfide removal both for the liquid and gas phase. The removal efficiency shows a decrease due to biofilm development, that shows an additional resistance to the sulfide reaction with the oxygen. Microorganisms related to the sulfide oxidation was found attached to the inner surface of the membrane, and the sample taken from the middle of membrane showed, in their majority, an anaerobic community. Therefore, in the future researches are required to find practical methods to reduce the biofilm inside the membrane by increasing the flow rate or by applying cleaning cycles.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This study was funded by the Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP), project process numbers 2013/13131-6, 2011/22904-3 and 2009/15984-0. David Jeison would like to thank the CRHIAM Centre (CONICYT/FONDAP/15130015) for their support and the Comisión Nacional de Investigación Científica y Tecnológica de Chile for their financial support and national doctoral scholarship.

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